Resolved Omitted-Shell Control for Regularized Determinants

A Quantitative Unit-Disk Tail Bound

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July 2026

Abstract

The reciprocal local-count audit does not stabilize, but the Arnoldi window contains additional complete shells beyond every selected RH-222 cloud. This paper asks whether those resolved omitted shells are small in the second-regularized determinant.

For $|w| < 1$,

$$|\log(1-w)+w| \le \frac{|w|^2}{2(1-|w|)}.$$

Consequently, if a finite omitted resonance multiset T satisfies $q = R\max_{\lambda\in T}|\lambda| < 1$, then on $|z| \le R$,

$$\left|\sum_{\lambda\in T}[\log(1-z\lambda)+z\lambda]\right| \le \frac{R^2}{2(1-q)}\sum_{\lambda\in T}|\lambda|^2.$$

This gives a branch-consistent uniform bound for the omitted $\det_2$ logarithm.

Each physical endpoint has 12–14 complete resolved roots beyond the selected cloud. On the closed unit disk, $q \le 0.29818$. The maximum analytic upper bound is 0.16804, the maximum observed 192-point grid tail is 0.07532, and every bound has positive slack.

The result controls the candidate-window tail only. It does not bound the unresolved infinite operator complement, and therefore does not establish a uniform small-noise determinant family.

1 Why second regularization helps

For a resonance multiset T, the finite second-regularized factor is

$$P_{T,2}(z) = \prod_{\lambda\in T}(1-z\lambda)e^{z\lambda}. \tag{1}$$

The exponential removes the linear term in the logarithm. Near $z = 0$, the tail is therefore quadratic in the omitted resonances rather than linear.

This is the finite-product counterpart of the Hilbert–Schmidt determinant used at fixed noise in RH-7 [1, 3]. It is exactly the regularization required when square summability is available but absolute summability is not.

2 The logarithmic tail theorem

Choose the logarithm by analytic continuation from zero on a disk where no factor vanishes.

Lemma 2.1 (One-factor estimate). *For* $|w| < 1$,

$$|\log(1-w)+w| \le \frac{|w|^2}{2(1-|w|)}. \tag{2}$$

Proof. The convergent power series gives

$$\log(1-w)+w=-\sum_{m=2}^{\infty}\frac{w^m}{m}.$$

Since $m^{-1}\le 1/2$ for $m\ge 2$,

$$\sum_{m=2}^{\infty}\frac{|w|^m}{m}\le\frac{1}{2}\sum_{m=2}^{\infty}|w|^m=\frac{|w|^2}{2(1-|w|)}.$$

□

Theorem 2.2 (Uniform finite $\det_2$ tail). *Let T be a finite resonance multiset and let*

$$q=R\max_{\lambda\in T}|\lambda|<1.$$

Then the logarithm of (1), normalized to vanish at zero, satisfies

$$\sup_{|z|\le R}|\log P_{T,2}(z)|\le B_R(T):=\frac{R^2}{2(1-q)}\sum_{\lambda\in T}|\lambda|^2. \tag{3}$$

Proof. Apply Lemma 2.1 to $w=z\lambda$. Since $|z\lambda|\le q$,

$$|\log(1-z\lambda)+z\lambda|\le\frac{|z|^2|\lambda|^2}{2(1-q)}.$$

Sum over T and use $|z|\le R$. □

Corollary 2.3 (Multiplicative tail). *On the same disk,*

$$|P_{T,2}(z)-1|\le e^{B_R(T)}-1.$$

Proof. Write $P_{T,2}=e^L$ and use $|e^L-1|\le e^{|L|}-1$. □

The logarithmic form is more informative when composing factors and comparing channels.

3 Resolved selected and omitted shells

At one endpoint, let

$$\Lambda_{\mathrm{cand}}=\Lambda_{\mathrm{sel}}\sqcup T_{\mathrm{res}}$$

be the complete-shell part of the Arnoldi candidate window, split after the RH-222 selected cloud. A nonreal candidate root whose conjugate lies outside the Arnoldi window is excluded from both sides; no synthetic partner is added.

Then

$$P_{\Lambda_{\mathrm{cand}},2}=P_{\Lambda_{\mathrm{sel}},2}P_{T_{\mathrm{res}},2}, \tag{4}$$

and Theorem 2.2 controls the exact ratio of the two finite products.

Each endpoint contains 12–14 roots in T_{res}. The count varies because real singleton shells and a possible incomplete candidate-boundary root alter parity.

4 Physical unit-disk audit

The disk radius is frozen at $R = 1$. The observed logarithmic tail is sampled at 48 angles on each of the radii

$$0.25, \quad 0.5, \quad 0.75, \quad 1.$$

The analytic bound itself holds at every point of the closed unit disk; the grid is used only to assess its slack.

diagnostic	result
endpoint count	32
resolved omitted roots	12–14
grid points per endpoint	192
maximum q	0.298172
maximum $\sum_{T_{\mathrm{res}}} \lvert\lambda\rvert^2$	archived
maximum logarithmic upper bound	0.168037
maximum observed grid log tail	0.075315
minimum bound slack	0.011053

Every omitted reciprocal zero has modulus at least $q^{-1} > 3.35$ when q is taken at its worst endpoint, so the unit disk is safely zero-free for every resolved omitted factor.

5 What the result does not bound

The full bulk spectrum decomposes as

$$\Lambda_{\mathrm{bulk}} = \Lambda_{\mathrm{sel}} \sqcup T_{\mathrm{res}} \sqcup T_{\mathrm{unres}}.$$

Theorem 2.2 controls T_{res} because its roots are explicitly stored. It supplies no value for

$$\sum_{\lambda \in T_{\mathrm{unres}}} |\lambda|^2.$$

At fixed noise, Hilbert–Schmidt theory guarantees that the infinite sum is finite and the truncations converge locally uniformly [1, 3]. The missing requirement is a bound uniform as $\sigma \downarrow 0$ after the chosen moving factors have been removed.

A whole-matrix Frobenius norm gives one formal upper bound on the unresolved eigenvalue square sum. The next paper tests it. Because the operators are nonnormal, that bound sees singular-direction mass that may be invisible to the resonance product and can be extremely pessimistic.

6 Claim boundary

The exact advance is a quantitative omitted-shell theorem and its successful application to every resolved candidate window. It does not certify Arnoldi roots, control the unresolved complement, or prove a small-noise normal family. The failed local-count gate of RH-227 is not repaired merely by a small resolved tail [2]. Gate A remains open.

References

[1] Barry Simon. *Trace Ideals and Their Applications.* American Mathematical Society, second edition, 2005.

[2] Liang Wang. A reciprocal local-count gate for the small-noise limit. 2026. RH-227 preprint.

[3] Liang Wang. Irreversible gaussian cycle spectrum. 2026. RH-7 preprint.